

Fleets of servers power Amazon Web Services (AWS). Senior management has requested data-driven solutions to optimize server usage.

Write a query that calculates the total time that the fleet of servers was running. The output should be in units of **full days**.

Assumptions:

- Each server might start and stop several times.
- The total time in which the server fleet is running can be calculated as the sum of each server's uptime.

`server_utilization` Table:

Column Name	Type
server_id	integer
status_time	timestamp
session_status	string

`server_utilization` Example Input:

server_id	status_time	session_status
1	08/02/2022 10:00:00	start
1	08/04/2022 10:00:00	stop
2	08/17/2022 10:00:00	start
2	08/24/2022 10:00:00	stop

Example Output:

total_uptime_days
21

Repeated the same mistake

```
server_id,  
min(status_time) start,  
max(status_time) as stop  
FROM server_utilization  
GROUP BY 1
```

Accurate Solution

```
WITH revised_server_view AS (  
select  
server_id,  
session_status,  
status_time,  
LAG(session_status) over(partition by server_id order by status_time) as prev_status,  
LAG(status_time) over(partition by server_id order by status_time) as prev_status_time  
from server_utilization )  
select  
SUM(datediff(status_time, prev_status_time)) as total_uptime_days  
from revised_server_view  
where prev_status = 'start'
```

server_id	session_status	status_time	prev_status	prev_status_time
1	start	2022-08-02 10:00:00	NULL	NULL
1	stop	2022-08-04 10:00:00	start	2022-08-02 10:00:00
1	start	2022-08-13 10:00:00	stop	2022-08-04 10:00:00
1	stop	2022-08-13 19:00:00	start	2022-08-13 10:00:00
2	start	2022-08-17 10:00:00	NULL	NULL
2	stop	2022-08-24 10:00:00	start	2022-08-17 10:00:00
3	start	2022-08-06 10:00:00	NULL	NULL
3	stop	2022-08-14 10:00:00	start	2022-08-06 10:00:00

server_id	session_status	status_time	prev_status	prev_status_time
1	stop	2022-08-04 10:00:00	start	2022-08-02 10:00:00
1	stop	2022-08-13 19:00:00	start	2022-08-13 10:00:00
2	stop	2022-08-24 10:00:00	start	2022-08-17 10:00:00
3	stop	2022-08-14 10:00:00	start	2022-08-06 10:00:00
3	stop	2022-08-19 10:00:00	start	2022-08-18 10:00:00
4	stop	2022-08-19 14:00:00	start	2022-08-16 10:00:00
5	stop	2022-08-19 10:00:00	start	2022-08-14 21:00:00